

AEM Training - Day 7 Assignment (26-03-2025)

1. Create `SampleServlet` Extending `SlingAllMethodsServlet`

Description:

AEM Servlet registered using `resourceType`.

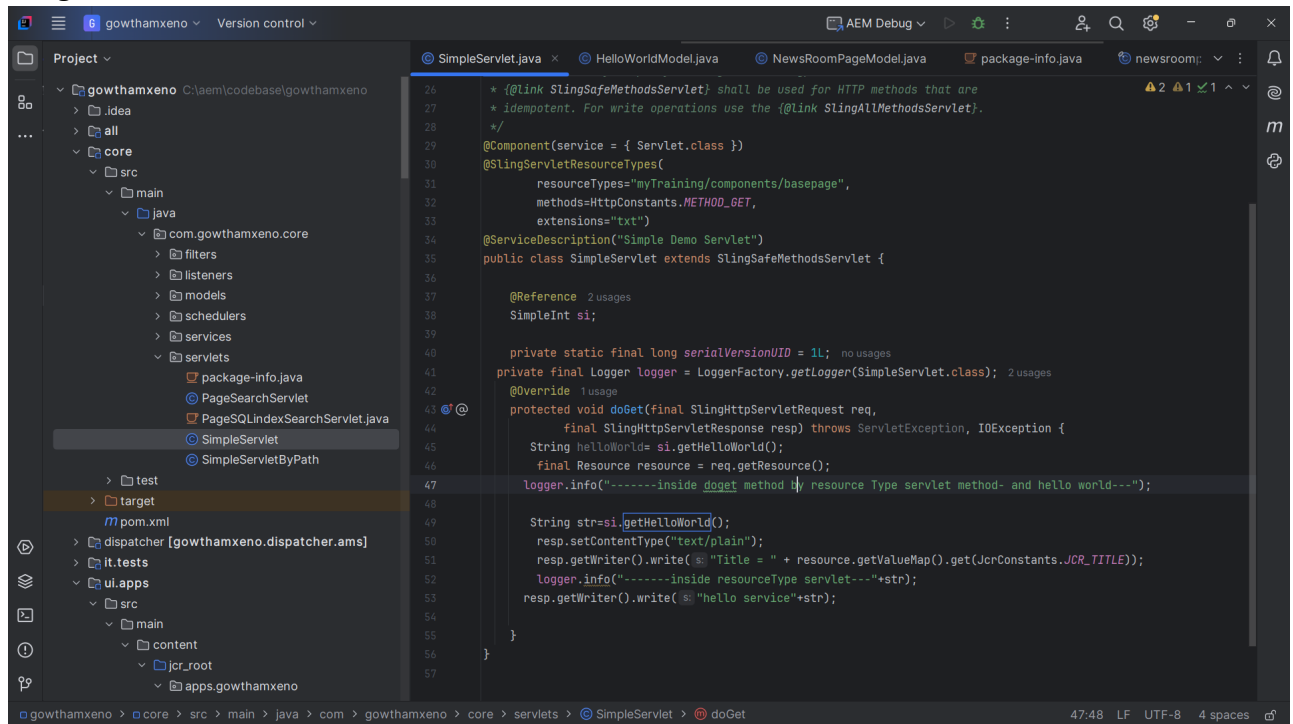
Code Snippet (`SampleServlet.java`):

```
@Component(service = Servlet.class, property = {
    "sling.servlet.resourceTypes=myapp/components/page",
    "sling.servlet.methods=GET"
})
public class SampleServlet extends SlingAllMethodsServlet {
    private static final Logger LOG =
        LoggerFactory.getLogger(SampleServlet.class);

    @Override
    protected void doGet(SlingHttpServletRequest request, SlingHttpServletResponse
response) throws ServletException, IOException {
        response.getWriter().write("Sample Servlet Executed");
        LOG.info("Sample Servlet Executed");
    }
}
```

Screenshots

1. SlingAllMethodsServlet



2. Create CreatePageServlet Extending SlingSafeMethodsServlet

Description:

A Servlet registered using **path** that creates pages based on user input.

Code Snippet (CreatePageServlet.java):

```

@Component(service = Servlet.class, property = {
    "sling.servlet.paths=/bin/createPage",
    "sling.servlet.methods=POST"
})
public class CreatePageServlet extends SlingSafeMethodsServlet {
    @Reference
    private ResourceResolverFactory resolverFactory;

    @Override
    protected void doPost(SlingHttpServletRequest request,
        SlingHttpServletResponse response) throws ServletException, IOException {
        String pageName = request.getParameter("pageName");
        try (ResourceResolver resolver =
            resolverFactory.getServiceResourceResolver(null)) {
            PageManager pageManager = resolver.adaptTo(PageManager.class);
            if (pageManager != null) {
                Page newPage = pageManager.create("/content/mysite", pageName,
                    "/apps/myapp/templates/page", pageName);
                resolver.commit();
                response.getWriter().write("Page created successfully: " +
                    newPage.getPath());
            } else {

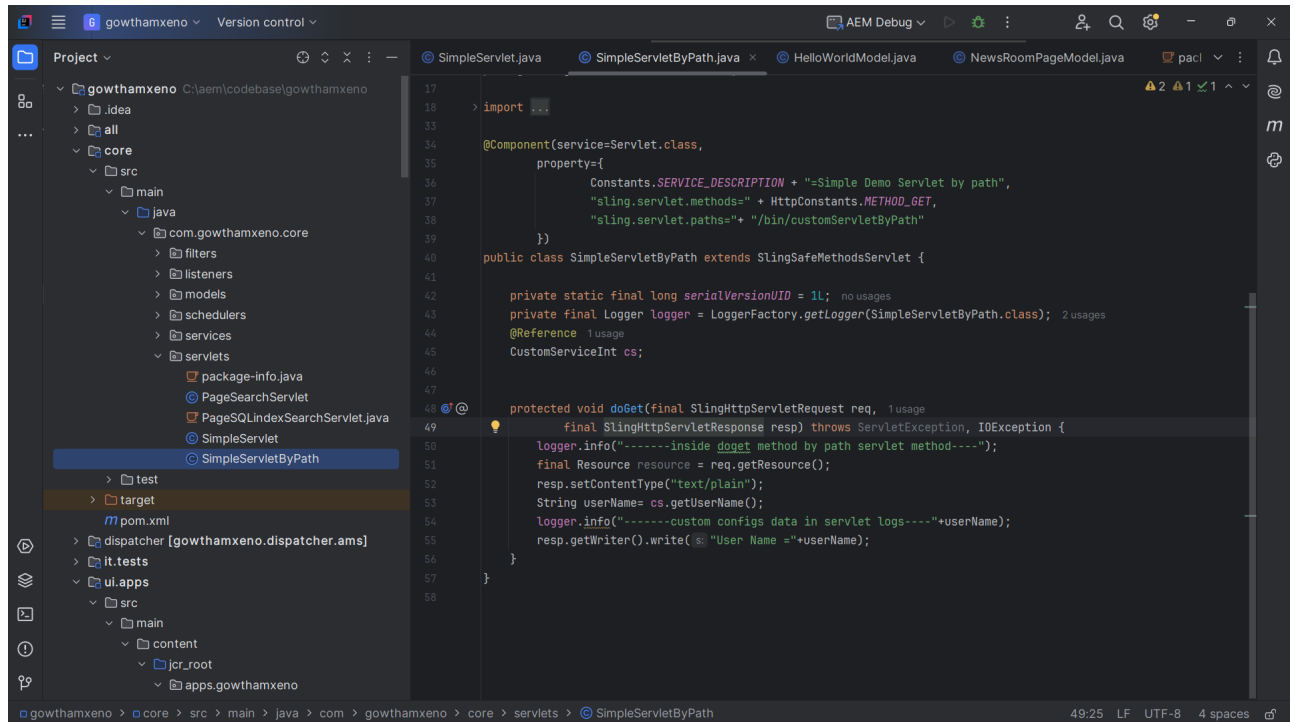
```

```

        response.getWriter().write("PageManager is null");
    }
} catch (PersistenceException e) {
    response.getWriter().write("Page creation failed: " + e.getMessage());
}
}
}

```

1. SlingSafeMethodsServlet



3. Use PageManager API to Create Pages

Description:

The above servlet uses the **PageManager API** to create pages dynamically.

Steps:

1. Get the **PageManager** instance from **ResourceResolver**.
2. Call the **create()** method with parameters: parent path, page name, template path, and title.
3. Commit the changes using **resolver.commit()**.

Code Snippet:

```

PageManager pageManager = resolver.adaptTo(PageManager.class);
if (pageManager != null) {
    Page newPage = pageManager.create("/content/mysite", pageName,
    "/apps/myapp/templates/page", pageName);
    resolver.commit();
}

```

4. Create `SearchServlet` Using `PredicateMap`

Description:

AEM Servlet to search content using **QueryBuilder API**.

Code Snippet (`SearchServlet.java`):

```
@Component(service = Servlet.class, property = {
    "sling.servlet.paths=/bin/searchContent",
    "sling.servlet.methods=GET"
})
public class SearchServlet extends SlingSafeMethodsServlet {
    @Reference
    private QueryBuilder queryBuilder;

    @Override
    protected void doGet(SlingHttpServletRequest request, SlingHttpServletResponse
response) throws ServletException, IOException {
        Map<String, String> map = new HashMap<>();
        map.put("path", "/content/mysite");
        map.put("type", "cq:Page");
        map.put("p.limit", "-1");

        Query query = queryBuilder.createQuery(PredicateGroup.create(map),
request.getResourceResolver().adaptTo(Session.class));
        SearchResult result = query.getResult();
        for (Hit hit : result.getHits()) {
            response.getWriter().write(hit.getPath() + "\n");
        }
    }
}
```